

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461047.874 +/- 0.05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.046
Transit depth (Rp/Rs)^2	2.52 +/- 1.45 [%]
Orbital Inclination (inc)	85.32 +/- 2.88
Airmass coefficient 1 (a1)	0.55 +/- 0.32
Airmass coefficient 2 (a2)	0.58 +/- 1.62
Scatter in the residuals of the lightcurve fit is	61.06 %
Best Comparison Star	#1 - [255, 198]
Optimal Aperture	3.82
Optimal Annulus	10.83
Transit Duration (day)	0.059 +/- 0.041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.159 ± 0.046 vs 0.1276 (deviation 0.68σ)
Duration fit vs published	0.059 d vs 0.1179 d (deviation 1.44σ)
Mid-time O–C	56.2 min
Depth SNR	3.5
Residual scatter	61.06 %

Light curve

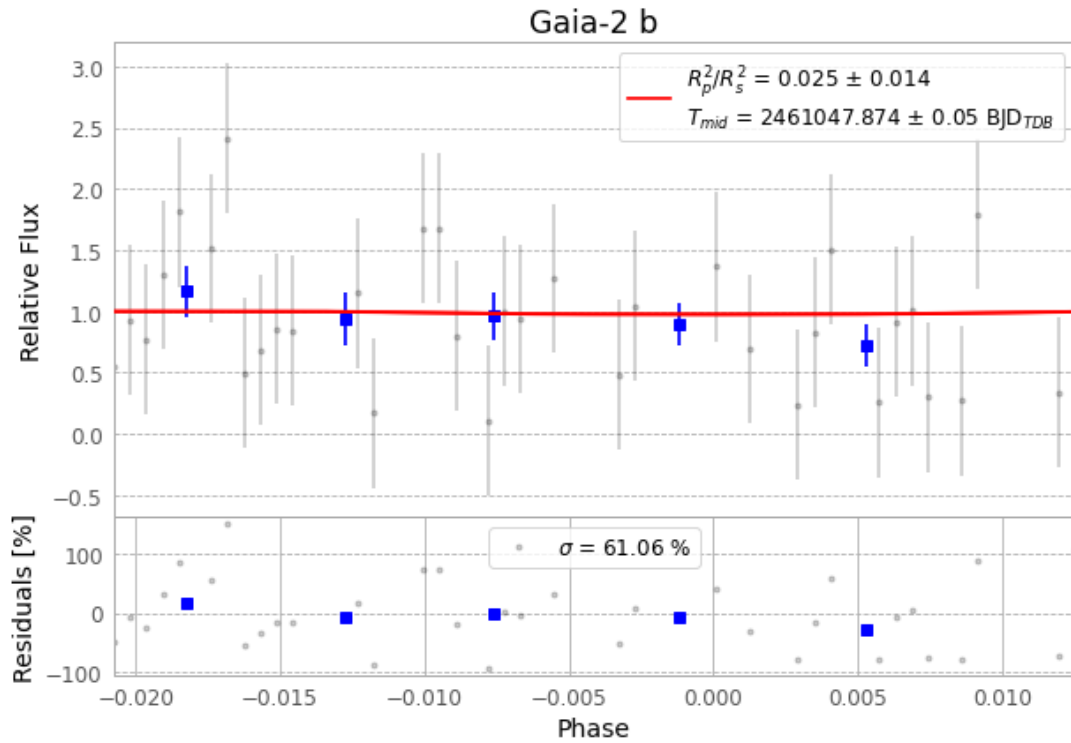


Figure 1 — FinalLightCurve_Gaia-2 b_2026-01-07.png (SHA-256 a69936c00b0a4b55...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [586, 357], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7151e7415e35b410...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

63 FITS frames (41 MB), Gaia-2260107070015.FITS → Gaia-2260107100615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Gaia-2-260107.log (c860f3a6657a...), FinalLightCurve_Gaia-2 b_2026-01-07.png (a69936c00b0a...), FinalLightCurve_Gaia-2 b_2026-01-07.csv (3384f97e6268...)

Compute resources

Wall time 218.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 22:51Z.